

Exact marginal crossover for a quartic Bose gas

A convergent expansion, rigorous computation,
and uniform lattice corrections

Research article and computational supplement

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Abstract. We sharpen the marginal logarithmic crossover formula of Łebek and Jakubczyk, *Physical Review A* **102** (2020), 013324, by evaluating its previously unspecified next constant and proving a convergent expansion with a sharp radius and an explicit remainder. For

$$\Psi(u) = \int_{\mathbb{R}} e^{-x^4 - ux^2} dx, \quad G(s) = \sum_{n \geq 1} \frac{\Psi(s\sqrt{n})}{n},$$

we establish

$$G(s) = \Gamma(\tfrac{1}{4}) \left(\log \frac{1}{s} + \frac{\pi}{4} + \frac{3}{2} \log 2 \right) + \frac{1}{2} \sum_{k \geq 1} \frac{(-1)^k \Gamma(\tfrac{1}{4} + \tfrac{k}{2}) \zeta(1 - \tfrac{k}{2})}{k!} s^k.$$

The correction series has radius exactly $\sqrt{8\pi}$ and converges absolutely on its convergence circle. In the published variable $\theta = s^2/8$, the constant next to $\log(1/\theta)$ is exactly $\pi/2$. A Mellin-contour estimate proves equality, including the absence of an unaccounted exponentially small term.

The expansion yields a summation algorithm requiring $O(\log(1/\varepsilon))$ special-function evaluations for absolute accuracy ε , uniformly over positive stiffness, when combined with an explicitly bounded large-stiffness tail. We derive a convergent Lambert- W expansion for the inverse critical temperature. Finally, for a specified finite-range lattice dispersion, we prove an all-orders low-temperature expansion uniform as the stiffness vanishes. Its first two lattice coefficients give corrections to the inverse critical line that dominate the exponentially small continuum stiffness corrections. The package includes arbitrary-precision reference code, outward-rounded Arb enclosures, independent quadrature, tests, and data.

Scope. The results are analytical theorems about explicitly defined Bose integrals and their lattice realization. The logarithmic crossover and its physical setting are prior work. This report provides a precise quantitative strengthening; it makes no claim to resolve a famous open conjecture or to establish the interacting Heisenberg model results in the inspirational repository. Publication priority beyond the sources reviewed has not been established. The proofs are not machine formalized.

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1 The research target and the precise advance

The starting inspiration was the heat-kernel approach to lattice corrections in Family 271 of the `openai/math` collection [9, 10]. In its elementary ideal-gas calculation, a uniform expansion of a lattice heat kernel is evaluated at integer multiples of inverse temperature and summed over those integers. We ask what remains true when the quadratic coefficient in a direction vanishes, so that the limiting dispersion becomes quartic and the Bose sum reaches its marginal integrability threshold.

This question leads to a concrete published target. Section IV.2 of Łebek–Jakubczyk [1], especially their equation (25), obtains the leading logarithmic critical line for one quadratic and two quartic directions, with a small positive quadratic stiffness in one of the quartic directions. Their cutoff argument leaves the next, nonlogarithmic coefficient unspecified. We replace that cutoff argument with an exact Mellin calculation and control the contour remainder.

The methods themselves are classical. Harmonic sums and their Mellin transforms are treated systematically in [2]; convergent exponential–Bessel expansions appear in [3], and numerical acceleration of related Schlömilch sums in [4]. Paris’s stated exponential– K family has positive exponential damping and a linked power/order parameter. The present sum has an independent $n^{-3/4}$ weight and the cancellation boundary $e^{n\theta} K_{1/4}(n\theta)$. We do not attribute novelty to the general Mellin method. The concrete additions here are the evaluated marginal constant, the complete specialized convergent expansion and its remainder, the inverse formulas, and a uniform transfer to a specified lattice model.

Question	Established starting point	Result proved here
Marginal continuum crossover	Leading $T \log(1/\theta)$ law in [1]	Exact constant $\pi/2$ and every correction coefficient
Nature of the expansion	Cutoff asymptotic estimate	Convergent series; sharp radius; explicit contour remainder
Numerical evaluation	A slowly convergent positive Bessel sum	Uniform logarithmic term count with rigorous tail control
Inverse critical line	Logarithmic suppression of temperature	Lambert- W approximation and convergent correction recursion
Actual lattice	Continuum low-energy description	All-orders asymptotic expansion uniform in the degenerating stiffness

Table 1: The scope of the strengthening. “Uniform logarithmic term count” counts special-function evaluations, not bit operations.

All subsequent results are proved from the displayed integral definitions. No unverified top-level assertion from `openai/math` is an input. The repository is cited for the research direction and the heat-kernel organization of the calculation.

2 Definitions, normalization, and the marginal obstruction

2.1 The quartic heat kernel and its harmonic sum

For $u \geq 0$, define

$$\Psi(u) = \int_{\mathbb{R}} \exp(-x^4 - ux^2) dx, \quad \Psi(0) = \frac{1}{2} \Gamma(\tfrac{1}{4}). \quad (2.1)$$

For $u > 0$, dropping the quartic term gives

$$0 < \Psi(u) \leq \sqrt{\pi} u^{-1/2}. \quad (2.2)$$

Consequently

$$G(s) = \sum_{n=1}^{\infty} n^{-1} \Psi(s\sqrt{n}), \quad s > 0, \quad (2.3)$$

converges, since it is at most $\sqrt{\pi}s^{-1/2}\zeta(5/4)$. It is strictly decreasing in s . At $s = 0$, each heat-kernel factor is constant, and the harmonic sum diverges. The small positive stiffness therefore regulates a real logarithmic singularity; it cannot be set to zero inside the original sum.

The standard quartic integral also gives

$$\Psi(u) = \frac{\sqrt{u}}{2} e^{u^2/8} K_{1/4}(u^2/8). \quad (2.4)$$

One may establish this from the integral representation of K by substitution, or verify that both sides solve the same differential equation in u and have the Gaussian large- u normalization. The heat-integral definition remains primary throughout the proofs.

2.2 An exact continuum density

We work in units with the three positive leading energy coefficients and Boltzmann's constant normalized to one. Define, for $T, \delta > 0$,

$$N_{\text{con}}(T, \delta) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{dx dy dz}{\exp((x^2 + y^4 + z^4 + \delta z^2)/T) - 1}. \quad (2.5)$$

Expanding the Bose denominator into its positive exponential series and using Tonelli's theorem gives

$$N_{\text{con}}(T, \delta) = C_* T G(s), \quad C_* := \frac{\Gamma(\frac{1}{4})}{16\pi^{5/2}}, \quad s := \frac{\delta}{\sqrt{T}}. \quad (2.6)$$

Indeed the x integral is $\sqrt{\pi}(T/n)^{1/2}$, the y integral is $\frac{1}{2}\Gamma(\frac{1}{4})(T/n)^{1/4}$, and the z integral is $(T/n)^{1/4}\Psi(s\sqrt{n})$. This explicitly checks every factor of two and π .

The weighted dimension at $\delta = 0$ is $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$. The zero-gap Bose integral is then exactly at its logarithmic integrability threshold. For $\delta > 0$, the last direction is quadratic at sufficiently low energy and the weighted dimension increases to $5/4$, making (2.5) finite.

2.3 Connection with the published scaling variable

Set

$$\theta = \frac{s^2}{8}, \quad S(\theta) = \sum_{n \geq 1} n^{-5/4} \sqrt{n\theta} e^{n\theta} K_{1/4}(n\theta). \quad (2.7)$$

The exact identity

$$S(\theta) = 2^{1/4} \theta^{1/4} G(\sqrt{8\theta}) \quad (2.8)$$

connects our normalization to that of [1]. For the general dispersion $t_0(k_1 a_0)^2 + t(k_2 a_0)^4 + \tau(k_3 a_0)^2 + \tau'(k_3 a_0)^4$, replace

$$s = \frac{\tau}{\sqrt{\tau' k_B T}}, \quad \theta = \frac{\tau^2}{8\tau' k_B T}, \quad (2.9)$$

and multiply the normalized density by $[a_0^3 \sqrt{t_0} t^{1/4} (\tau')^{1/4}]^{-1}$, using $k_B T$ for T . For the imperfect gas of [1], the critical chemical potential equals this ideal critical density times the mean-field coupling. These are changes of scale, not additional interaction theorems.

3 The exact convergent expansion

Put

$$R_* := \sqrt{8\pi}, \quad c_k := \frac{(-1)^k}{2k!} \Gamma\left(\frac{1}{4} + \frac{k}{2}\right) \zeta\left(1 - \frac{k}{2}\right) \quad (k \geq 1). \quad (3.1)$$

Theorem 3.1 (Exact expansion and a quantitative remainder). *For every real $0 < s \leq R_*$,*

$$G(s) = \Gamma\left(\frac{1}{4}\right) \left(\log \frac{1}{s} + \frac{\pi}{4} + \frac{3}{2} \log 2 \right) + \sum_{k=1}^{\infty} c_k s^k. \quad (3.2)$$

The correction series has radius exactly R_ as a complex power series and converges absolutely for every $|s| = R_*$. For $j \geq 1$, its partial sum through degree $4j + 1$ satisfies*

$$\left| G(s) - \Gamma\left(\frac{1}{4}\right) \left(\log \frac{1}{s} + \frac{\pi}{4} + \frac{3}{2} \log 2 \right) - \sum_{k=1}^{4j+1} c_k s^k \right| \leq B_j \left(\frac{s}{R_*} \right)^{4j+2}, \quad (3.3)$$

where

$$B_j = \frac{\sqrt{\pi} \zeta(2j+1)}{(4j+2)(2j+\frac{1}{4})^{1/4}}. \quad (3.4)$$

The same finite-contour inequality is valid for every $s > 0$, although its right side no longer tends to zero as $j \rightarrow \infty$ when $s > R_$.*

There is an exact zero pattern:

$$c_6 = c_{10} = c_{14} = \dots = 0. \quad (3.5)$$

The coefficient c_2 is not zero, because $\zeta(0) = -1/2$ is not a trivial zeta zero.

Corollary 3.2 (The missing constant). *With θ as in (2.7), for $0 < \theta \leq \pi$,*

$$S(\theta) = \frac{\Gamma(\frac{1}{4})}{2^{3/4}} \theta^{1/4} \left[\log \frac{1}{\theta} + \frac{\pi}{2} + \sum_{k \geq 1} d_k \theta^{k/2} \right], \quad (3.6)$$

where

$$d_k = \frac{(-1)^k 8^{k/2} \Gamma(\frac{1}{4} + \frac{k}{2}) \zeta(1 - \frac{k}{2})}{k! \Gamma(\frac{1}{4})}. \quad (3.7)$$

Thus the unspecified constant alongside the logarithm in the published marginal expression is exactly $\pi/2$.

For compact physical formulas, define

$$\mathcal{A} := \frac{\Gamma(\frac{1}{4})^2}{32\pi^{5/2}}, \quad b_k := \frac{2c_k}{\Gamma(\frac{1}{4})}, \quad B(s) := \sum_{k \geq 1} b_k s^k, \quad \kappa := 8e^{\pi/2}. \quad (3.8)$$

Then

$$\boxed{N_{\text{con}}(T, \delta) = \mathcal{A} T \left[\log \frac{8T}{\delta^2} + \frac{\pi}{2} + B(\delta/\sqrt{T}) \right]}. \quad (3.9)$$

This identity applies when $0 < \delta/\sqrt{T} \leq R_*$. The first terms are

$$\begin{aligned} b_1 &= -\frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \zeta\left(\frac{1}{2}\right) > 0, & b_2 &= -\frac{1}{16}, \\ b_3 &= -\frac{\Gamma(\frac{3}{4})}{8\Gamma(\frac{1}{4})} \zeta\left(-\frac{1}{2}\right), & b_4 &= -\frac{5}{4608}. \end{aligned} \quad (3.10)$$

In the θ variable the bracket is

$$\begin{aligned} \log \frac{1}{\theta} + \frac{\pi}{2} - 2\sqrt{2} \frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \zeta(\frac{1}{2}) \theta^{1/2} - \frac{1}{2} \theta \\ - 2\sqrt{2} \frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \zeta(-\frac{1}{2}) \theta^{3/2} - \frac{5}{72} \theta^2 + \dots \end{aligned} \quad (3.11)$$

The positive square-root correction is particularly easy to miss if a hard cutoff is substituted for the crossover kernel.

4 Proof: from a double pole to an exact identity

4.1 Mellin transform and its fundamental strip

For $0 < \Re z < 1/2$, absolute convergence permits interchanging integrals:

$$\begin{aligned} \int_0^\infty u^{z-1} \Psi(u) du &= \Gamma(z) \int_{\mathbb{R}} |x|^{-2z} e^{-x^4} dx \\ &= \frac{1}{2} \Gamma(z) \Gamma\left(\frac{1}{4} - \frac{z}{2}\right). \end{aligned} \quad (4.1)$$

The lower bound on $\Re z$ controls the integral at $u = 0$; the upper bound controls its algebraic $u^{-1/2}$ decay, obtained by Gaussian scaling at infinity. Scaling $u = s\sqrt{n}$ and summing produces

$$\mathcal{M}(z) := \int_0^\infty s^{z-1} G(s) ds = \frac{1}{2} \Gamma(z) \Gamma\left(\frac{1}{4} - \frac{z}{2}\right) \zeta\left(1 + \frac{z}{2}\right). \quad (4.2)$$

For a vertical line $\Re z = c$ inside this strip, the gamma factors decay exponentially as $e^{-3\pi|\Im z|/4}$ times a power. Thus Mellin inversion is absolutely integrable and gives

$$G(s) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \mathcal{M}(z) s^{-z} dz. \quad (4.3)$$

This is a concrete application of the harmonic-sum framework in [2]. The contour estimates below make the particular use of that framework self-contained.

4.2 The double pole and the exact finite constant

Write γ for Euler's constant and $\psi = \Gamma'/\Gamma$. At zero,

$$\begin{aligned} \Gamma(z) &= z^{-1} - \gamma + O(z), \\ \zeta(1 + z/2) &= 2z^{-1} + \gamma + O(z), \\ \Gamma(\tfrac{1}{4} - z/2) &= \Gamma(\tfrac{1}{4}) \left(1 - \tfrac{1}{2}\psi(\tfrac{1}{4})z + O(z^2)\right). \end{aligned}$$

Their product is

$$\mathcal{M}(z) = \Gamma(\tfrac{1}{4}) \left[\frac{1}{z^2} - \frac{\gamma + \psi(\frac{1}{4})}{2z} + O(1) \right]. \quad (4.4)$$

The gamma reflection and duplication identities [6] imply

$$\psi(\tfrac{1}{4}) = -\gamma - \frac{\pi}{2} - 3 \log 2. \quad (4.5)$$

For example, the digamma reflection formula gives $\psi(3/4) - \psi(1/4) = \pi$, while duplication gives $\psi(1/4) + \psi(3/4) = 2\psi(1/2) - 4\log 2$ and $\psi(1/2) = -\gamma - 2\log 2$. These relations yield (4.5) directly.

Multiplication by $s^{-z} = 1 - z\log s + O(z^2)$ shows that the residue at zero is precisely $\Gamma(\frac{1}{4})[-\log s + \pi/4 + (3/2)\log 2]$. Euler's constant cancels. The constant cannot be recovered reliably by fixing an arbitrary truncation index in the original positive sum, because the discarded crossover tail contributes at the same order.

At $z = -k$, $k \geq 1$, the residue of $\Gamma(z)$ is $(-1)^k/k!$. The other gamma factor is regular there, giving $c_k s^k$. When $k = 6, 10, 14, \dots$, the zeta zero cancels this gamma pole. All poles of $\Gamma(1/4 - z/2)$ lie to the right, at $z = 1/2 + 2j$, and are not crossed in a leftward displacement.

4.3 A far-left contour bound

A list of residues alone does not prove an exact convergent expansion: one must control the displaced contour. The zeta functional equation [7], followed by gamma reflection and duplication, gives the meromorphic identity

$$\mathcal{M}(z) = -\frac{\pi^{3/2}}{2} (8\pi)^{z/2} \frac{\Gamma(\frac{1}{4} - \frac{z}{2})}{\Gamma(\frac{1}{2} - \frac{z}{2})} \frac{\zeta(-z/2)}{z \sin(\pi z/4) \cos(\pi z/2)}. \quad (4.6)$$

This transformation exposes both the eventual radius R_* and the vertical exponential decay in a usable form.

Choose the line $\Re z = -4j - 2$, $j \geq 1$, and set $z = -4j - 2 + it$. The apparent pole at its real point is removable by the trivial zero already described. On this line,

$$\begin{aligned} |\sin(\pi z/4)| &= \cosh(\pi t/4), & |\cos(\pi z/2)| &= \cosh(\pi t/2), \\ |z| &\geq 4j + 2, & |\zeta(-z/2)| &\leq \zeta(2j + 1). \end{aligned} \quad (4.7)$$

To bound the gamma ratio, put $w = -z/2$. The beta integral yields

$$\left| \frac{\Gamma(w + \frac{1}{4})}{\Gamma(w + \frac{1}{2})} \right| \leq \frac{\Gamma(\Re w + \frac{1}{4})}{\Gamma(\Re w + \frac{1}{2})} \leq (\Re w - \frac{3}{4})^{-1/4}. \quad (4.8)$$

For the last inequality use

$$\frac{\Gamma(x + \frac{1}{4})}{\Gamma(x + \frac{1}{2})} = \frac{1}{\Gamma(\frac{1}{4})} \int_0^1 t^{x-3/4} (1-t)^{-3/4} dt,$$

bound $t^{x-3/4}$ by $\exp(-(x-3/4)(1-t))$, and extend the resulting integral in $1-t$ to $(0, \infty)$. Here $x = 2j + 1 > 3/4$.

The integral along the new vertical line is therefore bounded by

$$\frac{\sqrt{\pi}}{4} \frac{\zeta(2j+1)}{(4j+2)(2j+\frac{1}{4})^{1/4}} \left(\frac{s}{R_*} \right)^{4j+2} \int_{-\infty}^{\infty} \frac{dt}{\cosh(\pi t/4) \cosh(\pi t/2)}.$$

The remaining integral is at most $\int_{\mathbb{R}} \operatorname{sech}(\pi t/4) dt = 4$, proving (3.3). For each fixed j , the horizontal sides of the contour rectangle vanish as their height tends to infinity: gamma asymptotics and polynomial zeta growth on that fixed strip provide an integrable exponential majorant. This justifies every finite contour displacement before letting j grow.

If $0 < s \leq R_*$, the bound tends to zero. Thus the sum of residues is the function itself, even at the positive boundary point. In particular, there is no additional term flat to all algebraic orders at $s = 0$.

4.4 Sharp radius and a convenient coefficient bound

For $k \geq 3$, the zeta functional equation also gives

$$c_k = \frac{2\sqrt{\pi}}{k} (-1)^k \cos \frac{\pi k}{4} \frac{\Gamma(\frac{k}{2} + \frac{1}{4})}{\Gamma(\frac{k}{2} + \frac{1}{2})} \zeta(\frac{k}{2}) R_*^{-k}. \quad (4.9)$$

By (4.8),

$$|c_k| \leq K k^{-5/4} R_*^{-k}, \quad K := 2\sqrt{2\pi} \zeta(\frac{3}{2}). \quad (4.10)$$

This proves absolute convergence on the whole circle $|s| = R_*$. For every $M \geq 2$ and $r = |s|/R_* < 1$, it supplies the alternative tail bound

$$\left| \sum_{k>M} c_k s^k \right| \leq \frac{K}{(M+1)^{5/4}} \frac{r^{M+1}}{1-r}. \quad (4.11)$$

For $k = 4j$, the cosine has magnitude one, $\zeta(k/2) \rightarrow 1$, and the gamma ratio is asymptotic to $(k/2)^{-1/4}$. Hence $|c_{4j}|^{1/(4j)} \rightarrow R_*^{-1}$, proving that the radius is exactly R_* . This completes the proof of Theorem 3.1.

Remark 4.1 (Analyticity at the marginal endpoint). The function G itself diverges at zero. The regularized function $G(s) + \Gamma(\frac{1}{4}) \log s$ extends analytically to a disk of radius R_* . The radius statement concerns this regular part and the power series, not an ordinary Taylor series of G at its singular point. No location or type of every singularity on the boundary is asserted here.

5 Rigorous evaluation and the improvement in summation cost

5.1 Why direct positive summation is expensive

Let $P_N(s) = \sum_{n=1}^N n^{-1} \Psi(s\sqrt{n})$. From (2.2),

$$0 < G(s) - P_N(s) \leq 4\sqrt{\pi} s^{-1/2} N^{-1/4}. \quad (5.1)$$

For $u \geq 1$, integration over $|x| \leq u^{-1/2}$ gives $\Psi(u) \geq 2e^{-2} u^{-1/2}$. Therefore, if $s\sqrt{N+1} \geq 1$,

$$G(s) - P_N(s) \geq 8e^{-2} s^{-1/2} (N+1)^{-1/4}. \quad (5.2)$$

For fixed $s > 0$ as $\varepsilon \downarrow 0$, unaccelerated truncation consequently requires $\Theta(s^{-2}\varepsilon^{-4})$ terms to reach absolute error ε . The same scaling holds in a joint regime with $s^{-2}\varepsilon^{-4} \gg \max(1, s^{-2})$. This qualification matters: if s is sufficiently large relative to a fixed tolerance, even $G(s)$ can be smaller than that tolerance. These estimates describe direct positive truncation, not a lower bound on all possible algorithms for G .

5.2 Complementary large-stiffness control

The Gaussian scaling $x = y/\sqrt{u}$ in (2.1) gives

$$\Psi(u) = u^{-1/2} \int_{\mathbb{R}} e^{-y^2} e^{-y^4/u^2} dy.$$

Taylor's theorem for e^{-v} on $v \geq 0$ has a signed remainder bounded by its next term. Thus, for integers $M \geq 1$, $N \geq 0$,

$$G(s) = P_N(s) + \sum_{k=0}^{M-1} \frac{(-1)^k \Gamma(2k + \frac{1}{2})}{k! s^{2k+1/2}} \zeta(k + \frac{5}{4}, N+1) + E_{M,N}(s), \quad (5.3)$$

$$0 \leq (-1)^M E_{M,N}(s) \leq \frac{\Gamma(2M + \frac{1}{2})}{M! s^{2M+1/2}} \zeta(M + \frac{5}{4}, N+1). \quad (5.4)$$

Here $P_0 = 0$ and $\zeta(p, a)$ is the Hurwitz zeta function. Unlike (3.2), the infinite series in inverse powers of s is generally divergent; only its finite truncations and their signed bounds are used.

Theorem 5.1 (Uniform logarithmic evaluation count). *Assume arithmetic and elementary-function operations, and evaluations of gamma, Riemann and Hurwitz zeta, and scaled $K_{1/4}$, are counted at unit cost. For every $s > 0$ and $0 < \varepsilon < 1/2$, an approximation to $G(s)$ with analytical error at most ε can be computed using $O(\log(1/\varepsilon))$ such evaluations, with the implied constant independent of s .*

Proof. If $s \leq R_*/2$, use (3.3); a number of coefficients proportional to $\log(1/\varepsilon)$ suffices, uniformly in s . If $s > R_*/2$, choose

$$M \geq \max \left(1, \left\lceil \log_2 \frac{\sqrt{\pi}}{\varepsilon} \right\rceil \right), \quad N = \left\lceil \frac{4M}{s^2} \right\rceil.$$

The decreasing-series integral bound and a central-binomial estimate give

$$\zeta(M + \frac{5}{4}, N + 1) \leq \frac{N^{-M-1/4}}{M + 1/4}, \quad \frac{\Gamma(2M + \frac{1}{2})}{M!} \leq \sqrt{\pi}(2M)^M.$$

For the second inequality, use $\Gamma(2M + 1/2) = \sqrt{\pi}(4M)!/[4^{2M}(2M)!]$, then $\binom{4M}{2M} \leq 4^{2M}$ and $(2M)!/M! \leq (2M)^M$. Since $s^2 N \geq 4M$, the bound in (5.4) is at most

$$\frac{\sqrt{\pi} 2^{-M}}{(M + 1/4)(4M)^{1/4}} \leq \sqrt{\pi} 2^{-M} \leq \varepsilon.$$

Moreover $N \leq 2M/\pi + 1$, because $s^2 > 2\pi$. The positive prefix and the Hurwitz tail therefore use $O(M)$ terms. $\square$

The theorem is deliberately stated in a special-function operation model. It does not claim an $O(\log(1/\varepsilon))$ bit complexity: input encoding, working precision, evaluation of special functions, and the number of output digits must be charged in a bit model. Near $s = 0$ the magnitude of $G(s)$ grows like $\log(1/s)$, so even representing an absolute-accuracy result can require additional precision. The provided code exposes its precision and error information explicitly.

5.3 Floating-point checks and genuine enclosures

Two implementations are included. The `mpmath` reference evaluator returns a numerical approximation and the proved analytical truncation bound. That bound alone does not enclose rounding error. The separate `python-flint/Arb` implementation evaluates all constants, coefficients, and partial sums using balls and enlarges the result by an outward-rounded analytical remainder. Arb's semantics are documented in [11]. These resulting balls include both types of error.

The independent certified comparison uses (5.3) and scaled Bessel evaluations rather than the small- s Mellin coefficients. Agreement is checked by overlap of the enclosures. An additional independent integral formulation is

$$G(s) = \int_{\mathbb{R}} \text{Li}_{3/4}(e^{-x^4 - sx^2}) dx = 4 \int_0^\infty t \text{Li}_{3/4}(e^{-t^8 - st^4}) dt. \quad (5.5)$$

The first identity follows by substituting $x = n^{1/4}y$ into each term of (2.3), then applying Tonelli. The second removes the integrable square-root singularity at the origin. Its integrand tends to $4\Gamma(\frac{1}{4})s^{-1/4}$ at zero. Quadrature of this expression gives checks that do not reuse either coefficient expansion.

6 Inverting the continuum critical line

Fix a density $n_* > 0$. For each $\delta > 0$, the function $N_{\text{con}}(T, \delta)$ is continuous and strictly increasing in T . It tends to zero as $T \downarrow 0$ and to infinity as $T \rightarrow \infty$. There is therefore a unique T_{con} satisfying $N_{\text{con}}(T_{\text{con}}, \delta) = n_*$. We study $\delta \downarrow 0$ with n_* fixed.

Define

$$r_* := \frac{n_*}{\mathcal{A}}, \quad w := W_0\left(\frac{\kappa r_*}{\delta^2}\right), \quad T_0 := \frac{r_*}{w}, \quad D := w + 1. \quad (6.1)$$

The principal Lambert function W_0 satisfies $W_0(x)e^{W_0(x)} = x$ for $x > 0$ [5]. Consequently T_0 solves the full logarithm-plus-constant model exactly:

$$\mathcal{A}T_0 \log \frac{\kappa T_0}{\delta^2} = n_*. \quad (6.2)$$

It is preferable to retain this exact W expression rather than truncate its nested-logarithm expansion prematurely.

Theorem 6.1 (First two continuum inverse corrections). *As $\delta \downarrow 0$ at fixed $n_* > 0$,*

$$T_{\text{con}} = T_0 - \frac{b_1 \delta \sqrt{T_0}}{D} + \delta^2 \left(\frac{1}{16D} + \frac{b_1^2 w}{2D^3} \right) + O\left(\frac{\delta^3}{\sqrt{T_0} D} \right). \quad (6.3)$$

More generally, the relative correction $u = T_{\text{con}}/T_0 - 1$ has a convergent expansion in $z = \delta/\sqrt{T_0}$ whose coefficients are explicitly recursive rational expressions in w and the b_k . There is a positive radius for this local inverse independent of $w \geq 0$, and, on a smaller disk, the remainder after degree m is $O_m(|z|^{m+1}/(w+1))$ uniformly in w .

Proof. Substitute $T = T_0(1 + u)$ into (3.9), subtract (6.2), and divide by $\mathcal{A}T_0$. The exact equation is

$$Du + E(u) + \sum_{k \geq 1} b_k z^k (1 + u)^{1-k/2} = 0, \quad (6.4)$$

where

$$E(u) = (1 + u) \log(1 + u) - u = \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} u^j. \quad (6.5)$$

All powers use their branch analytic near $u = 0$.

For a fully quantitative-in-order justification, choose a sufficiently small fixed disk $|u| \leq r < 1$ so that $\sup |E'(u)| < 1/2$. Since $B(v)$ is analytic and $B(0) = 0$, both $B(z/\sqrt{1+u})$ and its contribution to the u derivative tend to zero uniformly on this disk as $z \rightarrow 0$. Thus the map

$$u \mapsto -\frac{E(u) + (1 + u)B(z/\sqrt{1+u})}{D}$$

maps the disk into itself and is a strict contraction for $|z| < z_0$, where $z_0 > 0$ is independent of $D \geq 1$. Its analytic iterates converge uniformly on smaller disks, giving an analytic solution. The equation also gives $|u| \leq C|z|/D$. Cauchy estimates on a fixed smaller z circle imply the stated uniform Taylor remainder.

Writing $u = \sum_{m \geq 1} a_m(w) z^m$, coefficient comparison gives

$$\begin{aligned} a_1(w) &= -\frac{b_1}{D}, \\ a_2(w) &= \frac{1}{16D} + \frac{b_1^2 w}{2D^3}. \end{aligned} \quad (6.6)$$

For arbitrary m , let $U_{m-1}(z) = \sum_{j < m} a_j(w) z^j$. The recursion is

$$a_m(w) = -\frac{1}{D}[z^m] \left\{ E(U_{m-1}) + \sum_{k=1}^m b_k z^k (1 + U_{m-1})^{1-k/2} \right\}. \quad (6.7)$$

Only previously computed coefficients occur on the right. Multiplying the first two terms and the $O(z^3/D)$ remainder by T_0 proves (6.3). Uniqueness of the physical critical temperature identifies this local inverse with T_{con} for small δ . $\square$

6.1 Nested logarithms and exponentially small sectors

If $L = \log(\kappa r_*/\delta^2)$ and $\ell = \log L$, the classical Lambert expansion [5] gives

$$w = L - \ell + \frac{\ell}{L} + \frac{\ell(\ell-2)}{2L^2} + O\left(\frac{\ell^3}{L^3}\right). \quad (6.8)$$

In particular,

$$T_{\text{con}} \sim \frac{n_*}{2\mathcal{A} \log(1/\delta)}. \quad (6.9)$$

The correction variable has the useful exact identity

$$z^2 = \frac{\delta^2}{T_0} = \kappa e^{-w}. \quad (6.10)$$

Thus the Lambert expression already sums the entire hierarchy of nested logarithms, while (6.7) adds a convergent series in $e^{-w/2}$ with rational-in- w coefficients. This is a concrete asymptotic expansion involving both logarithmic and exponential scales. The convergence claimed here is local in z , not a blanket convergence claim about every rearrangement into powers of $1/L$.

6.2 Global conditioning and a one-evaluation temperature enclosure

Proposition 6.2 (Global elasticity bounds). *For every $T, \delta > 0$, set $s = \delta/\sqrt{T}$ and $\beta(s) = -sG'(s)/G(s)$. Then*

$$0 < \beta(s) < \frac{1}{2}, \quad 1 < \frac{\partial \log N_{\text{con}}}{\partial \log T} < \frac{5}{4}. \quad (6.11)$$

At fixed density, the critical temperature consequently satisfies

$$0 < \frac{\partial \log T_{\text{con}}}{\partial \log \delta} < \frac{2}{5}, \quad \frac{4}{5} < \frac{\partial \log T_{\text{con}}}{\partial \log n_*} < 1. \quad (6.12)$$

In particular, for $a > 1$, $T_{\text{con}}(\delta) < T_{\text{con}}(a\delta) < a^{2/5} T_{\text{con}}(\delta)$.

Proof. For $u > 0$, integration of the derivative of $x \exp(-x^4 - ux^2)$ gives

$$\Psi(u) = 4 \int_{\mathbb{R}} x^4 e^{-x^4 - ux^2} dx + 2u \int_{\mathbb{R}} x^2 e^{-x^4 - ux^2} dx.$$

Since $-\Psi'(u)$ equals the second-moment integral,

$$-\frac{u\Psi'(u)}{\Psi(u)} = \frac{1}{2} - 2 \frac{\int_{\mathbb{R}} x^4 e^{-x^4 - ux^2} dx}{\Psi(u)} \in (0, 1/2).$$

Gaussian moment bounds justify differentiating the defining series for G uniformly on compact positive s intervals. The logarithmic derivative $\beta(s)$ is a positive weighted average of the displayed kernel ratios, with weights proportional to $n^{-1}\Psi(s\sqrt{n})$. Hence $0 < \beta(s) < 1/2$.

The exact scaling identity $N_{\text{con}} = C_* T G(s)$ now gives

$$\frac{\partial \log N_{\text{con}}}{\partial \log T} = 1 + \frac{\beta(s)}{2}, \quad \frac{\partial \log N_{\text{con}}}{\partial \log \delta} = -\beta(s).$$

Implicit differentiation at the unique critical temperature yields $\partial \log T_{\text{con}} / \partial \log \delta = \beta / (1 + \beta/2)$ and $\partial \log T_{\text{con}} / \partial \log n_* = 1 / (1 + \beta/2)$. Their stated bounds follow, and integration of the stiffness derivative proves the multiplicative comparison. $\square$

Corollary 6.3 (Density-to-temperature enclosure). *Fix a trial temperature $T > 0$, the target density $n_* > 0$, and the stiffness $\delta > 0$. Suppose a rigorous density evaluation gives*

$$0 < \ell \leq \frac{N_{\text{con}}(T, \delta)}{n_*} \leq h < \infty.$$

Then

$$T_{\text{con}} \in T \left[\min\{h^{-1}, h^{-4/5}\}, \max\{\ell^{-1}, \ell^{-4/5}\} \right]. \quad (6.13)$$

Thus a single density enclosure converts an arbitrary positive temperature approximation into a rigorous temperature enclosure.

Proof. Integrating (6.11) between T and the exact critical temperature shows, with $r = N_{\text{con}}(T, \delta)/n_*$, that

$$\min\{r^{-1}, r^{-4/5}\} \leq \frac{T_{\text{con}}}{T} \leq \max\{r^{-1}, r^{-4/5}\}.$$

Both bounding functions are decreasing in $r > 0$. Replacing r by h in the lower bound and by ℓ in the upper bound proves the claim. A numerical implementation must outward round this final conversion as well as the density evaluation; ordinary floating-point substitution alone is not a certified enclosure. $\square$

7 An actual lattice and an exact change of variables

We now specify a lattice rather than assuming that a continuum dispersion remains accurate at all momenta. Let

$$q(k) := 2(1 - \cos k), \quad E_\delta(k) := q(k_1) + q(k_2)^2 + q(k_3)^2 + \delta q(k_3). \quad (7.1)$$

This is a nonnegative trigonometric polynomial and hence a finite-range lattice dispersion. The squared terms involve competing hopping coefficients; no assertion that all real-space couplings are ferromagnetic is made. For $T, \delta > 0$, define

$$N_{\text{lat}}(T, \delta) := \frac{1}{(2\pi)^3} \int_{[-\pi, \pi]^3} \frac{dk}{e^{E_\delta(k)/T} - 1}. \quad (7.2)$$

The exact monotone substitution

$$x_i = 2 \sin(k_i/2), \quad q(k_i) = x_i^2 \quad (7.3)$$

transforms the dispersion into $x_1^2 + x_2^4 + x_3^4 + \delta x_3^2$ on $[-2, 2]^3$. The entire lattice effect is now its domain and the explicit Jacobian

$$J(x) = \prod_{i=1}^3 (1 - x_i^2/4)^{-1/2}. \quad (7.4)$$

This identity is more useful here than a dispersion Taylor expansion: it preserves the stiff-to-soft crossover exactly before any asymptotic truncation is performed.

7.1 Weighted moments and coefficients

Let

$$\alpha_j := \frac{\binom{2j}{j}}{16^j} \quad (j \geq 0), \quad (7.5)$$

so that $(1 - x^2/4)^{-1/2} = \sum_{j \geq 0} \alpha_j x^{2j}$ for $|x| < 2$. For nonnegative integers a, b, c with $r = 2a + b + c > 0$, put

$$M_{abc}(s) := \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{u^{2a} v^{2b} w^{2c} du dv dw}{e^{u^2+v^4+w^4+sw^2} - 1}, \quad s \geq 0, \quad (7.6)$$

and

$$C_r(s) := \sum_{2a+b+c=r} \alpha_a \alpha_b \alpha_c M_{abc}(s), \quad r \geq 1. \quad (7.7)$$

These moments are finite, continuous on $[0, \infty)$, and nonincreasing in s . Their finiteness follows from the exact endpoint evaluation

$$M_{abc}(0) = \frac{\Gamma(a + \frac{1}{2}) \Gamma(\frac{b}{2} + \frac{1}{4}) \Gamma(\frac{c}{2} + \frac{1}{4})}{32\pi^3} \zeta\left(1 + \frac{2a + b + c}{2}\right). \quad (7.8)$$

To prove it, expand the positive Bose series and evaluate the three gamma integrals. The zeta argument exceeds one precisely because $2a + b + c > 0$. Domination by the $s = 0$ integrand then proves continuity and the bound $0 \leq M_{abc}(s) \leq M_{abc}(0)$.

Theorem 7.1 (All-orders uniform lattice transfer). *For every fixed integer $L \geq 0$, as $T \downarrow 0$,*

$$N_{\text{lat}}(T, \delta) = N_{\text{con}}(T, \delta) + T \sum_{r=1}^L T^{r/2} C_r(\delta/\sqrt{T}) + O_L(T^{1+(L+1)/2}), \quad (7.9)$$

uniformly for every $\delta > 0$. In particular,

$$|N_{\text{lat}}(T, \delta) - N_{\text{con}}(T, \delta)| \leq CT^{3/2} \quad (7.10)$$

for $0 < T \leq T_$, with constants independent of δ .*

Proof. Let $Q = [-1, 1]^3$. Outside Q , the polynomial $E_0(x) = x_1^2 + x_2^4 + x_3^4$ is at least one. Since $E_\delta \geq E_0$ and J is integrable on $[-2, 2]^3$, the lattice integral outside Q is exponentially small uniformly in δ . For the continuum tails, including any fixed polynomial weight, use $e^{-E_0/T} \leq e^{-1/(2T)} e^{-E_0/(2T)}$ on this complement. This supplies the same uniform exponential control.

On Q , expand the positive analytic Jacobian series

$$J(x) = \sum_{a,b,c \geq 0} \alpha_a \alpha_b \alpha_c x_1^{2a} x_2^{2b} x_3^{2c}$$

and retain all terms of weighted degree $2a + b + c \leq L$. The remaining series is bounded by a constant depending on L times the sum of finitely many monomials of degrees $L + 1$ and $L + 2$. Here is a direct justification. From any omitted multi-index, subtract coordinates one at a time until the weighted degree first reaches the retained range; just before that step its degree is $L + 1$ or $L + 2$, because the coordinate weights are $2, 1, 1$. Thus every omitted monomial contains such a boundary monomial as a factor. Assign it to one boundary monomial. The sum of the remaining factors and their coefficients is bounded on Q , because the untruncated series is absolutely convergent there. This proves the stated finite polynomial majorant.

Scale $x_1 = \sqrt{T}u$ and $x_2 = T^{1/4}v$, $x_3 = T^{1/4}w$. The measure contributes T , and the multi-index (a, b, c) contributes $T^{a+(b+c)/2}$. Extending the resulting integrals back to all of $\mathbb{R}^3$ costs

only the already bounded exponential tails. The constant term is N_{con} ; every positive-degree term is exactly $T^{1+(2a+b+c)/2}M_{abc}(\delta/\sqrt{T})$.

All remainder moments are bounded by their finite $s = 0$ values in (7.8). For $T \leq 1$, both boundary degrees therefore give at most a constant times $T^{1+(L+1)/2}$. This proves (7.9) uniformly in $\delta > 0$. $\square$

Remark 7.2 (The value at zero stiffness). At $\delta = 0$, both individual densities are infinite. The difference nevertheless has a finite common-cutoff limit: on the central cube the integrand is $(J-1)/(e^{E_0/T} - 1)$, and the remaining pieces are finite tails. The theorem concerns ordinary finite densities for $\delta > 0$. Whenever a numerical table below uses $s = 0$, it evaluates this regularized difference, never an undefined subtraction $\infty - \infty$.

8 Explicit lattice corrections and the joint scale structure

Let $\rho_{\Gamma} := \Gamma(\frac{3}{4})/\Gamma(\frac{1}{4})$. The first two coefficients are

$$C_1(s) = \frac{1}{8}(M_{010}(s) + M_{001}(s)), \quad (8.1)$$

$$C_2(s) = \frac{1}{8}M_{100}(s) + \frac{3}{128}(M_{020}(s) + M_{002}(s)) + \frac{1}{64}M_{011}(s). \quad (8.2)$$

At the marginal endpoint,

$$\mathcal{B} := C_1(0) = \frac{\sqrt{2}\zeta(\frac{3}{2})}{128\pi^{3/2}} = 0.005183412791031944\dots, \quad (8.3)$$

$$\mathcal{D} := C_2(0) = \mathcal{A}\zeta(2) \left(\frac{19}{256} + \frac{\rho_{\Gamma}^2}{64} \right) = 0.002935761063925785\dots \quad (8.4)$$

For (8.3), use $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$. In (8.4), the 19/256 combines the quadratic-direction Jacobian term with the two pure quartic-direction fourth moments; the remaining term comes from their mixed second moments.

Theorem 7.1 immediately gives, if $T \rightarrow 0$ and $s = \delta/\sqrt{T} \rightarrow 0$,

$$N_{\text{lat}} - N_{\text{con}} = \mathcal{B}T^{3/2} + o(T^{3/2}). \quad (8.5)$$

The convergence $s \rightarrow 0$ alone does not justify replacing the last little- o by $O(T^2)$.

Proposition 8.1 (A modulus at the marginal point). *For $0 < s \leq 1$,*

$$C_1(s) = \mathcal{B} + O(s(1 + |\log s|)), \quad C_2(s) = \mathcal{D} + O(s). \quad (8.6)$$

Consequently, for $0 < \delta/\sqrt{T} \leq 1$,

$$N_{\text{lat}} - N_{\text{con}} = \mathcal{B}T^{3/2} + O\left(T^2 + \delta T(1 + |\log(\delta/\sqrt{T})|)\right). \quad (8.7)$$

Proof. Write $\Psi_c(u) = \int_{\mathbb{R}} x^{2c} e^{-x^4 - ux^2} dx$. The Bose series gives

$$M_{abc}(s) = \frac{\Gamma(a + \frac{1}{2})\Gamma(\frac{b}{2} + \frac{1}{4})}{16\pi^3} \sum_{n \geq 1} n^{-1-(2a+b+c)/2} \Psi_c(s\sqrt{n}). \quad (8.8)$$

Dropping either the quartic or quadratic exponential proves $|\Psi'_c(u)| \leq C_c \min(1, u^{-c-3/2})$. For degree one, differentiation leaves a sum bounded by a constant times $\sum n^{-1} \min(1, (s\sqrt{n})^{-c-3/2})$. Splitting at $n = s^{-2}$ bounds it by $C(1 + |\log s|)$. Integrating from zero to s proves the first assertion. For degree two, the differentiated series already has summable exponent 3/2 at $s = 0$, proving the second. Equation (8.7) follows from the $L = 1$ lattice expansion. $\square$

8.1 Why logarithms reappear in the coefficient functions

The behavior in Proposition 8.1 is structural. For integers $r, c \geq 0$, the sum

$$F_{r,c}(s) = \sum_{n \geq 1} n^{-1-r/2} \Psi_c(s\sqrt{n})$$

has Mellin transform

$$\frac{1}{2} \Gamma(z) \Gamma\left(\frac{2c+1}{4} - \frac{z}{2}\right) \zeta\left(1 + \frac{r}{2} + \frac{z}{2}\right). \quad (8.9)$$

Its zeta pole lies at $z = -r$, exactly at a gamma pole. The resulting term at order s^r is

$$\frac{(-1)^r \Gamma(\frac{2c+1}{4} + \frac{r}{2})}{r!} s^r \left[-\log s + H_r - \frac{\gamma + \psi(\frac{2c+1}{4} + \frac{r}{2})}{2} \right], \quad (8.10)$$

where $H_0 = 0$. This follows by the same finite residue calculation as (4.4). Earlier orders have ordinary power coefficients. In particular, degree-one lattice corrections first acquire an $s \log s$ term. Equation (8.10) records a local residue consequence; a complete global analytic-continuation theorem for every $F_{r,c}$ is not needed for the uniform lattice theorem.

There are two different expansions in this problem. The continuum correction series in s is convergent. The full lattice expansion in $\sqrt{T}$ is an all-orders asymptotic expansion; its gamma moments generally grow too rapidly for convergence. A complete joint description also contains stiffness logarithms in the coefficient functions. Keeping these distinctions prevents invalid rearrangement of the two limiting processes.

9 The inverse critical temperature on the lattice

For fixed $n_* > 0$, define T_{lat} by $N_{\text{lat}}(T_{\text{lat}}, \delta) = n_*$. Existence and uniqueness follow from continuity, strict increase in T , the zero-temperature limit, and divergence as $T \rightarrow \infty$. Retain w, T_0 from (6.1).

Theorem 9.1 (First two lattice inverse corrections). *As $\delta \downarrow 0$ at fixed $n_* > 0$,*

$$T_{\text{lat}} = T_0 - \frac{\mathcal{B}T_0^{3/2} + \mathcal{D}T_0^2}{\mathcal{A}(w+1)} + O(T_0^{7/2}). \quad (9.1)$$

The two displayed shifts have orders $w^{-5/2}$ and w^{-3} , respectively. The first omitted term is $O(w^{-7/2})$.

Proof. Uniformly on a fixed relative neighborhood of T_0 , the variable $\delta/\sqrt{T}$ is exponentially small in w , by (6.10). Equations (3.9), (7.9), and (8.6) then give

$$N_{\text{lat}}(T, \delta) = F_0(T) + \mathcal{B}T^{3/2} + \mathcal{D}T^2 + O(T^{5/2}), \quad F_0(T) = \mathcal{A}T \log(\kappa T/\delta^2), \quad (9.2)$$

with constants allowed to depend on the fixed density. The exponentially small stiffness terms are smaller than every power of T_0 . Monotonicity and comparison at $(1 \pm \eta)T_0$ first imply $T_{\text{lat}}/T_0 \rightarrow 1$.

At T_0 , $F_0(T_0) = n_*$ and $F'_0(T_0) = \mathcal{A}(w+1)$. Set

$$T_* = T_0 - \frac{\mathcal{B}T_0^{3/2} + \mathcal{D}T_0^2}{\mathcal{A}(w+1)}.$$

The shift is $O(T_0^{5/2})$ because $w = n_*/(\mathcal{A}T_0)$. Taylor expansion of the explicit F_0 gives a second-order residual $O(T_0^4)$, while changing the argument of the $\mathcal{B}T^{3/2}$ term contributes $O(T_0^3)$. Both

are smaller than the $O(T_0^{5/2})$ density remainder already present in (9.2). Thus $N_{\text{lat}}(T_*, \delta) - n_* = O(T_0^{5/2})$.

For a sufficiently large fixed K , evaluate (9.2) at $T_* \pm KT_0^{7/2}$. The explicit logarithmic term changes by

$$\pm K\mathcal{A}(w+1)T_0^{7/2}(1+o(1)) = \pm Kn_*T_0^{5/2}(1+o(1)),$$

which dominates the known density remainder when K is large enough. The exact density is therefore below n_* at the lower endpoint and above it at the upper endpoint. Strict monotonicity proves (9.1). This argument never differentiates the unspecified $O(T^{5/2})$ remainder. $\square$

The continuum correction in (6.3) is exponentially small in w after the logarithmic model has been inverted. In contrast, the first relative lattice correction is of order $w^{-3/2}$. Thus a continuum calculation with many powers of δ still omits a larger lattice term. For the actual lattice model, the hierarchy should be organized as follows: retain the Lambert function; expand lattice corrections in half-integer powers of $1/w$; and, when needed, add the smaller $e^{-w/2}$ stiffness sectors. The all-orders theorem gives the data necessary to continue this recursion beyond the displayed terms.

10 Computational evidence and reproducibility

The proofs above do not depend on numerical fitting. Computation serves three purposes: testing normalization against independent formulas, checking the explicit error estimates, and assessing how quickly the new terms improve a usable approximation. The data files preserve the precision and error category of each computation.

10.1 Certified continuum values

s	Retained degree	$G(s)$, rounded for display
10^{-100}	5	841.4447020599820234520051
10^{-12}	5	106.7964732238546136755949
10^{-6}	5	56.70682215252535178343695
10^{-3}	9	31.66288993282315111533207
0.1	17	15.05380448328100861789823
1	41	7.412762878290256239836740
3	129	4.606413185798615193856927
4.9	2685	3.646966730634974892134411

Table 2: Arb calculations at 90 decimal digits with requested enclosure radius 10^{-30} . The full balls, including rounding and truncation, are stored in `data/certified_validation.json`. Degrees refer to the small- s evaluator alone; the combined algorithm avoids its slow convergence near R_* .

At $s = 1, 3, 4.9$, independent enclosures obtained from a positive Bessel prefix and a signed Hurwitz-zeta tail overlap the Mellin enclosures. Their radii are required to be below 10^{-35} ; in the recorded run all three are below 6×10^{-57} . A width requirement is essential: an excessively wide interval can overlap another interval without providing a useful independent check. The combined evaluator also checks positive stiffness values beyond the Mellin convergence disk. The integral in (5.5) supplies a second kind of independent check at seven parameter values, including R_* . The boundary test deliberately uses a finite degree: it confirms the stated bound without disguising the slower convergence there.

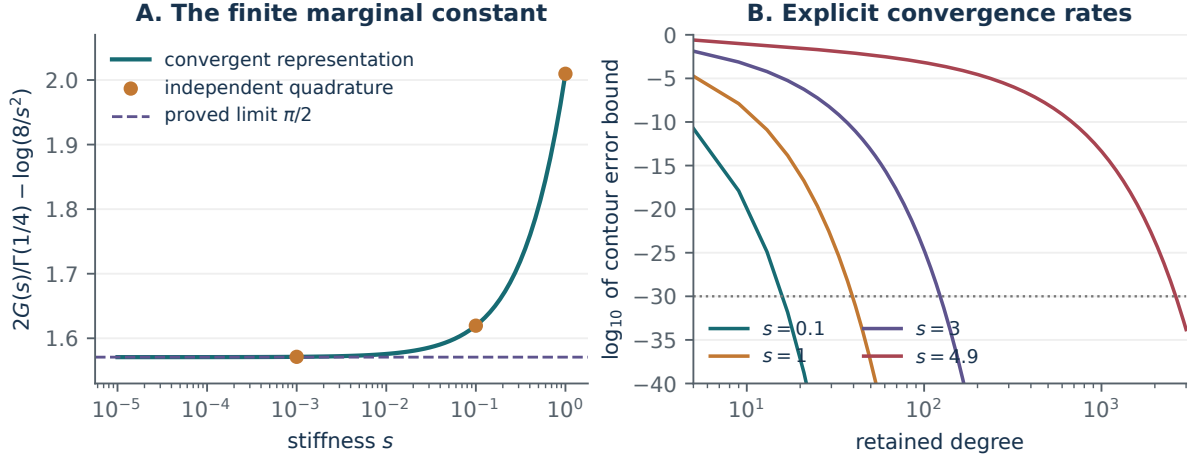


Figure 1: Left: the normalized finite part tends to $\pi/2$; the markers are independent polylogarithm quadrature. Right: the proved contour bound shows rapid convergence inside the disk and slower convergence near its boundary. Curves are evaluations of the analytical formulas, not fitted error models.

The test suite contains 42 passing checks covering independent representations, signed tails, exact coefficient zeros, inverse residuals, enclosure widths, and failure at insufficient precision. The symbolic recursion is checked exactly through degree six. A root certificate produced using Corollary 6.3 for $n_* = 0.1$ and $\delta = 0.01$ gives

$$T_{\text{con}} = 0.3595007501283353098918576878323848735202951682 \quad \text{within } 2 \times 10^{-46}.$$

The full interval and its density calculation are saved in the certified data. This uncertainty includes arithmetic rounding and the analytical tail; it is not an estimate based only on a solver residual.

10.2 Uniform lattice checks

The lattice validator integrates the already subtracted $J - 1$ expression in energy-radial and Dirichlet coordinates. This avoids subtracting two large densities near the marginal limit. It records an explicit bound on omitted large-energy tails, and assesses quadrature sensitivity by changing the quadrature order. The latter is a convergence diagnostic, not an interval proof of quadrature error.

At $s = 0$, understood as the regularized difference, the results are:

T	$(N_{\text{lat}} - N_{\text{con}})/T^{3/2}$	$((N_{\text{lat}} - N_{\text{con}})/T^{3/2} - \mathcal{B})/\sqrt{T}$
0.10	0.00615630028566	0.00307654039
0.05	0.00585920488505	0.00302223412
0.02	0.00560536047131	0.00298362066
0.01	0.00548013858365	0.00296725793
Predicted limit	$\mathcal{B} = 0.00518341279103$	$\mathcal{D} = 0.00293576106393$

Table 3: The first two uniform lattice coefficients. All displayed digits are numerical diagnostics; consult the data file for the coarse-fine differences and omitted-tail bounds.

Crossovers at $s = 0.01, 0.1, 1, 10$ are also included. The largest scaled coarse-fine discrepancy in those recorded tests is approximately 3.24×10^{-7} . The explicit omitted-tail bound in the test setup is below 2.4×10^{-16} ; these are separate error quantities.

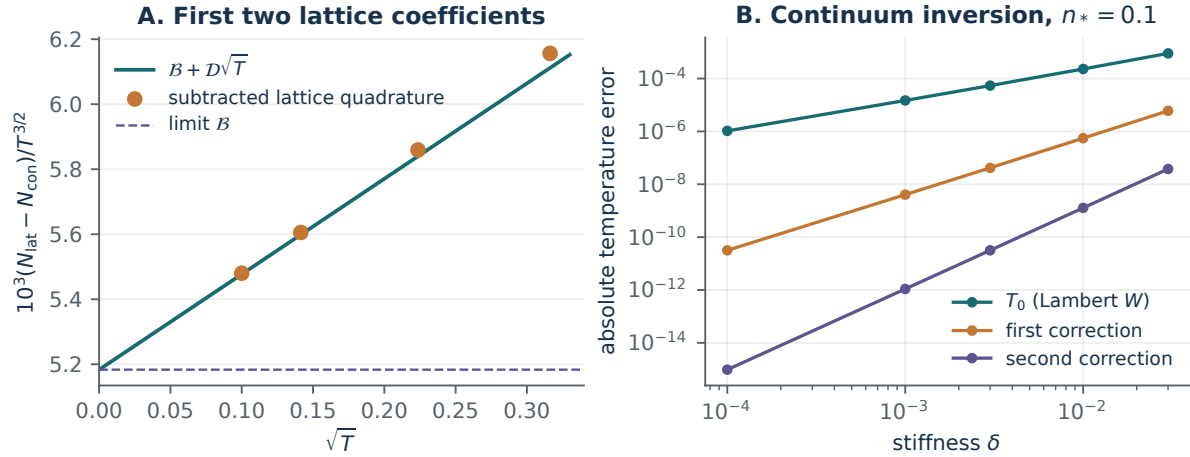


Figure 2: Left: the regularized lattice difference approaches its predicted two-term expansion. Right: the continuum inverse improves when the first and second stiffness corrections are included. These panels concern different asymptotic scales; the lattice theorem explains their ordering.

10.3 How to reproduce the package

The package root contains the article, code, tests, generated figures, and machine-readable data. The reference implementation requires `mpmath`; symbolic coefficient generation uses `sympy`; the lattice quadrature uses `numpy` and `scipy`; plots use `matplotlib`; the certified evaluator uses `python-flint 0.9.0`. Exact environment versions are recorded in the supplied environment report.

From the package root, the principal commands are:

```
python3 -m pytest -q
python3 code/bose_certified.py 0.1 --atol 1e-30
python3 code/bose_certified.py --validation data/certified_validation.json
python3 code/validate_independent.py
python3 code/lattice_validate.py
python3 code/make_figures.py
cd article
latexmk -xelatex -interaction=nonstopmode marginal_bose_crossover.tex
```

The independent quadrature is intentionally slower than the evaluator. The README provides the exact commands and optional arguments for each script. Numerical checks never replace the proof of a contour displacement, the sharp radius, or the uniform lattice remainder.

11 Proof audit, limitations, and interpretation

What the result strengthens. Equation (25) of [1] gives the correct leading logarithm. We do not identify that statement as erroneous. The improvement is the evaluation of the next coefficient and replacement of the leading approximation by an exact, effectively convergent representation. The explicit error estimate and the lattice theorem then support further quantitative conclusions.

The dangerous interchange of limits. Substituting $\Psi(0)$ in every summand makes the harmonic series diverge. Conversely, applying the large-argument Bessel asymptotic to every

term when s is small misses the transition region. The Mellin transform handles that region without a freely chosen cutoff. In the lattice proof, positive-degree moments eliminate the marginal divergence from the remainder; this is why its constant is uniform in stiffness.

A convergent series versus a formal residue list. The far-left estimate (3.3) is essential. A formal residue calculation could produce a convergent-looking series while leaving an unexamined contour term. Here the contour term is bounded explicitly and tends to zero on the stated interval. The separate coefficient estimate proves the exact complex radius.

Numerical certification. An analytical remainder bound combined with ordinary floating-point evaluation is not automatically a certified interval. The package therefore distinguishes its `mpmath` output from its Arb balls. Quadrature refinement is reported as numerical evidence. Only the enclosure implementation claims to include rounding in its interval. A concrete numerical issue arose in the reference implementation: very small Hurwitz-zeta values can lose relative accuracy before multiplication by large gamma factors. The code evaluates $\zeta(p, a)$ with an additional $\lceil p \log_{10} a \rceil + 20$ digits and includes a regression check. The Arb implementation uses the proved adaptive prefix allocation and rejects requested accuracies unless the final ball radius meets the tolerance. In particular, overlap alone is never accepted as an independent certificate.

Model dependence. The continuum constant pertains to the explicitly normalized crossover integral. The coefficients $\mathcal{B}$ and $\mathcal{D}$ pertain to (7.1). A different lattice with the same leading exponents can have different higher-order coefficients. The physical mean-field interpretation does not prove the same formulas for a gas with arbitrary short-range interactions, nor for the quantum Heisenberg ferromagnet. Those questions require additional estimates controlling interactions.

Priority and research status. The literature review located the precise leading law being strengthened and related Mellin–Bessel technology. It did not locate the complete specialized statement proved here. That is a bounded literature finding, not a proof of publication priority. The package is a research draft with analytical proofs and independent computational checks, suitable for further mathematical review. No Lean or other kernel-checked formalization is included.

12 Further research questions

The following questions follow directly from the established formulas. They are arranged so that the first few can be attacked without introducing an interacting many-body theory.

1. **Locate and classify the nearest complex singularities.** The coefficient theorem proves the sharp radius R_* . Determine the singularity type at each point on that circle and construct analytic continuation across the remaining arcs. The late coefficient factor $\cos(\pi k/4)$ gives specific directional information, but a complete singularity theorem requires more than the radius calculation.
2. **Develop a joint expansion of every lattice coefficient.** Use (8.9) to obtain exact expansions of $C_r(s)$, including the $s^r \log s$ terms. Establish coefficient bounds uniform in r sufficient to control a joint limit in T and s , rather than only each fixed order of the lattice expansion.
3. **Higher-order soft directions.** Replace $x^4 + sx^2$ by $x^{2m} + \sum_{j < m} s_j x^{2j}$ and place the remaining directions at a marginal weighted dimension. Determine finite constants, resonance

logarithms, and convergence regions for several stiffness parameters. The relevant critical points of the polynomial suggest a geometric description of the convergence boundary.

4. **Nonseparable quartic forms.** Study, for example, $x^2 + y^4 + z^4 + \eta y^2 z^2 + \delta z^2$ with a positive quartic form. Identify which parts of the marginal constant have a closed form, and prove uniformity as the quartic form itself approaches degeneracy. The separable gamma products used here will need replacement by controlled angular integrals.
5. **A complete bit-complexity theorem.** Charge the cost of gamma, zeta, logarithm, scaled Bessel evaluation, and input/output precision. Investigate whether all necessary half-integer zeta values can be generated together efficiently. The special-function count in Theorem 5.1 is a useful starting point, but is not this stronger theorem.
6. **Certified lattice inversion and optimal evaluation near R_* .** The continuum density-to-temperature certificate is already supplied by Corollary 6.3. Extend this approach to T_{lat} using certified quadrature. Compare the simple two-regime scheme with rational approximants or contour deformations. Any improved method should preserve an explicit global error certificate.
7. **Finite-size crossover.** Introduce finite periodic lengths in the soft directions and track the joint limits of temperature, stiffness, and volume. Poisson summation can connect the present heat-kernel approach to the Casimir questions in [1, 8]; the zero mode must be kept separate from the thermodynamic density.
8. **Universality under controlled lattice perturbations.** Replace the exact sine-coordinate model by a class of analytic nonnegative lattice symbols sharing the same leading weighted jet. State and prove conditions under which the logarithmic constant is unchanged and compute how the first lattice correction changes. Uniform control of the complement of the minimum is necessary.
9. **Interactions beyond the mean-field model.** Determine whether a rigorously bounded self-energy or effective dispersion approximation preserves the marginal constant or only the leading logarithm. A useful first target would be an explicit bound comparing the interacting density to the ideal density on the scale of $T^{3/2}$, with stiffness dependence stated rather than suppressed.
10. **Formal verification of the analytic core.** Separate a potential formalization into the positive-integral identities, Mellin inversion and contour displacement, gamma/zeta identities, and the explicit majorants. Formalizing only the algebra of the residues would leave the crucial equality and uniformity questions unverified; the contour and domination arguments should be explicit goals.

A Supplementary coefficient and inverse identities

The coefficient formula (4.9) is preferable numerically for $k \geq 3$ because it uses zeta only at real arguments greater than one. The eight-periodic cosine should be handled algebraically: when $k \equiv 2 \pmod{4}$ the coefficient is exactly zero, subject to the separate $k = 2$ exception. Computing a tiny approximate cosine at a mathematical zero would introduce spurious coefficients.

For example, the coefficient of s^4 in the normalized bracket is

$$b_4 = \frac{\Gamma(9/4)}{24\Gamma(1/4)} \zeta(-1) = \frac{5}{384} \left(-\frac{1}{12}\right) = -\frac{5}{4608}.$$

After multiplication by 8^2 , this becomes $d_4 = -5/72$ in (3.11). This cross-check distinguishes the two common normalizations used in the report.

The recursion (6.7) also gives, with $D = w + 1$,

$$a_3 = -\frac{1}{D} \left[a_1 a_2 - \frac{a_1^3}{6} + b_1 \left(\frac{a_2}{2} - \frac{a_1^2}{8} \right) + b_3 \right]. \quad (\text{A.1})$$

The $b_2 z^2$ term has no u dependence, because its exponent $1 - k/2$ is zero. This removes a possible source of an incorrect second or third inverse coefficient. The accompanying symbolic generator derives further coefficients from the exact implicit equation and verifies their residuals.

B Source provenance and artifact notes

The repository was inspected at commit

`fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

The directly relevant source is Family 271, particularly the section “The explicit lattice correction” of *The first lattice correction to Bloch’s law* [10]. The article’s independent mathematical target is the marginal continuum formula in [1]. Sources were checked on 8 October 2026. The source manifest records URLs and roles; no third-party manuscript is repackaged as an original contribution.

The archive is self-contained for rebuilding this article once the listed Python and LaTeX dependencies are installed. Data and figures are generated artifacts; the analytical statements can be read and checked without executing them. The PDF has been compiled and visually reviewed. The complete validation outcome and the environment are recorded separately so that the tests can be rerun after integration into another repository.

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